

Advanced Database Models Exercise 1

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Immatriculation Number: 256245

Question 1

SNo.	System	Conform	Reason
1	Spreadsheet Applications (e.g. Microsoft Excel)	✗	Spreadsheet Applications are not Databases, because they don't support transactions and don't guarantee integrity control
1	Network File Systems	✗	These also don't support transactions
1	MySQL	✓	Yes, because they support all 9 rules of Codd
1	IBM DB2	✓	Yes, because they support all 9 rules of Codd

Question 2

Advantages of logical data independence

- application is unaffected by changes to the schema
- adding new columns or tables doesn't break the system

Advantages of physical data independence

- changing storage location doesn't break the system
- changing storage platform doesn't break the system
- distributing the data doesn't affect the system (too much)

Question 3

```
Student(matrNr, firstName, lastName, birthDate)
Lecture(id, title, lecturerName)
Attends(matrNr, id, semester, grade)
```

The following possible integrity constraints can be expressed for them:

- if a Student attends a lecture then his `matrNr` should be present in both `Student` and `Attends` tables
- every `Lecture` should have at least one Student attending
- `matrNr` for every `Student` must be unique
- `id` for every `Lecture` must be unique